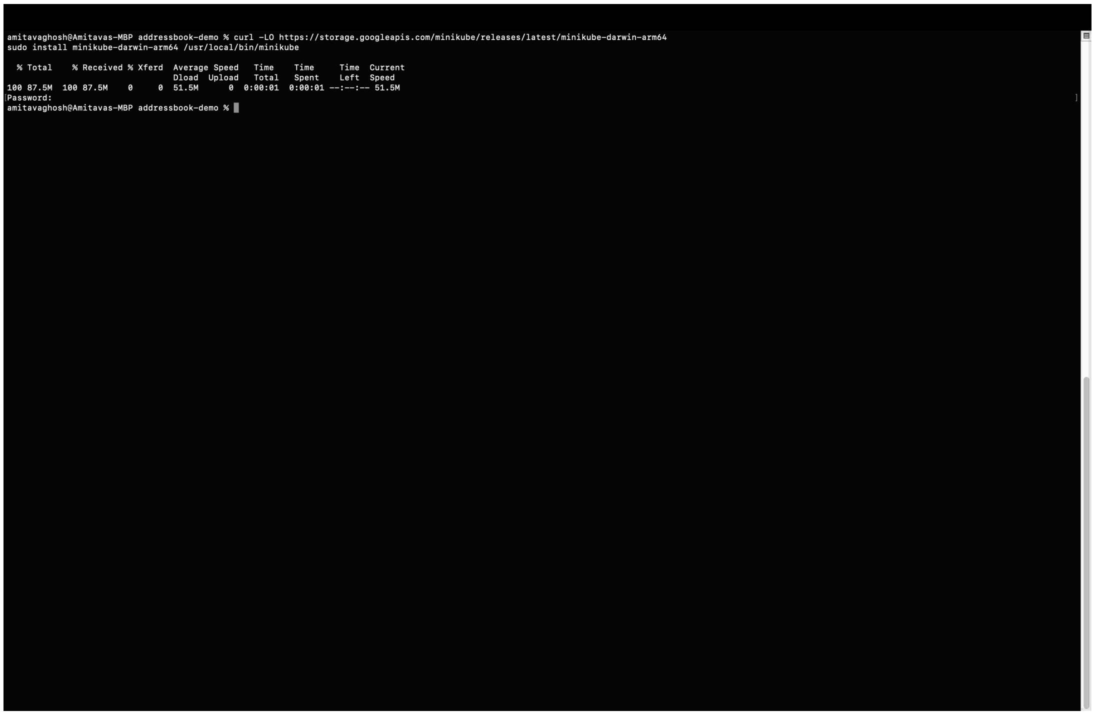
Kubernetes Assignment Solution

1. Install minikube on a machine.

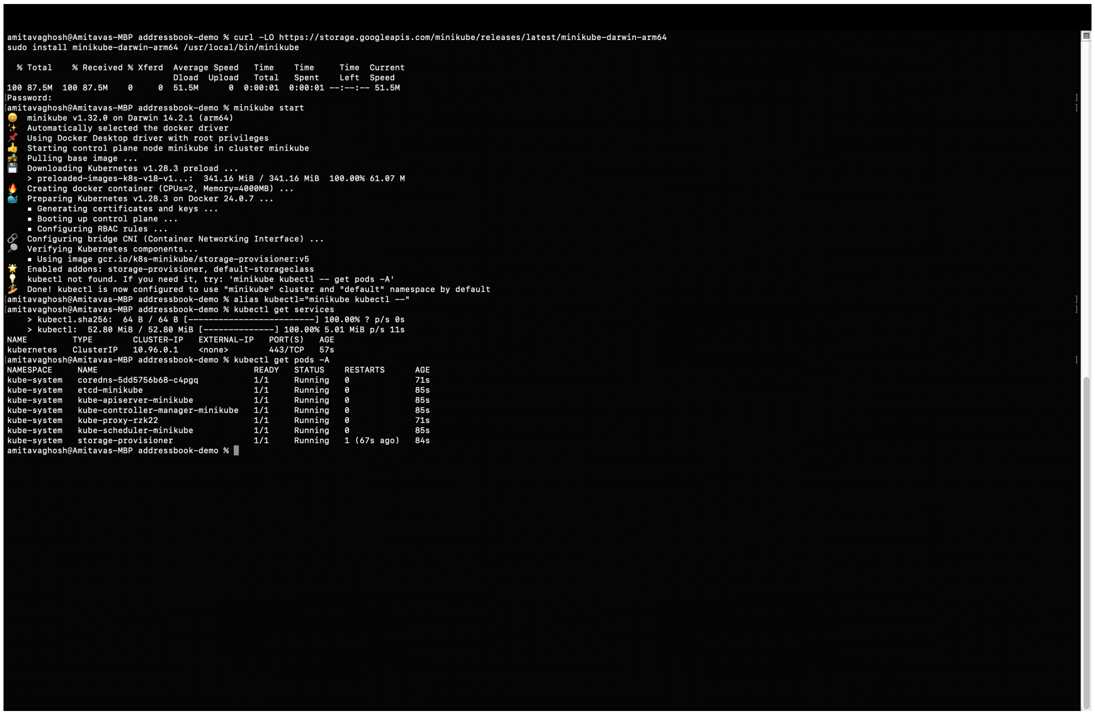


2. Start minikube cluster.

minikube start

Creating alias for **minikube kubectl —**

alias kubectl="minikube kubectl --"
kubectl get pods -A



```
amitavaghosh@Amitavas-MBP addressbook-demo % curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-darwin-arm64
sudo install minikube-darwin-arm64 /usr/local/bin/minikube

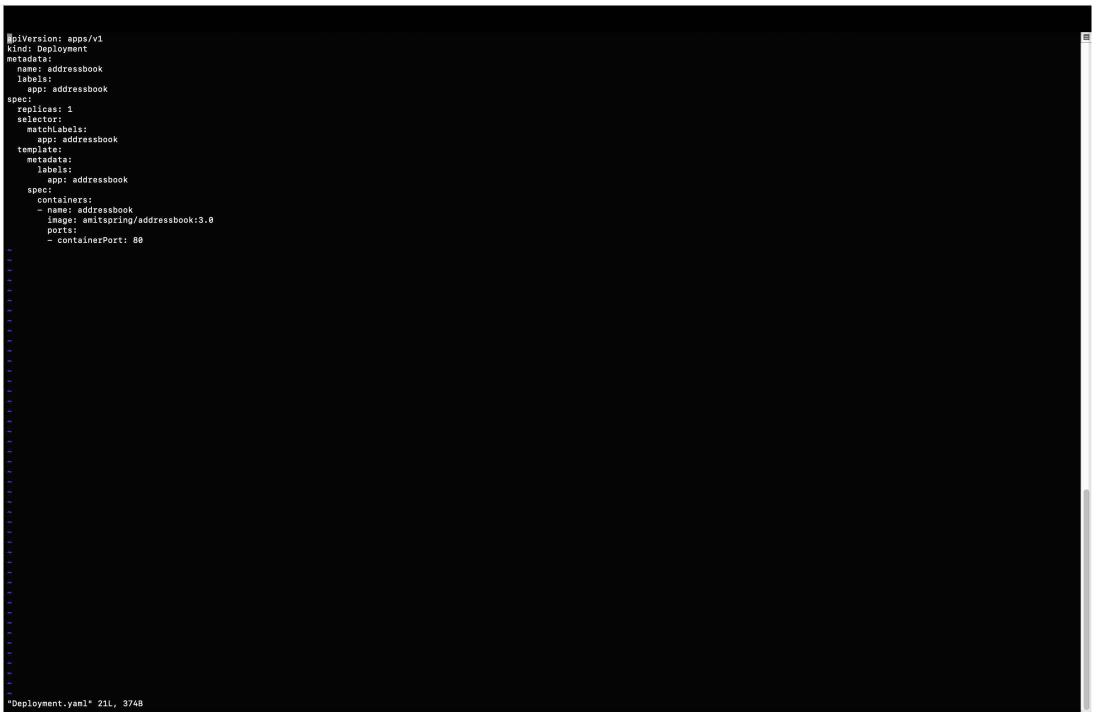
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 87.5M  100 87.5M    0     0  51.5M      0  0:00:01  0:00:01 --:--:-- 51.5M
Password:
amitavaghosh@Amitavas-MBP addressbook-demo % minikube start
😄  minikube v1.32.0 on Darwin 14.2.1 (arm64)
✨  Automatically selected the docker driver
📌  Using Docker Desktop driver with root privileges
👍  Starting control plane node minikube in cluster minikube
🚜  Pulling base image ...
💾  Downloading Kubernetes v1.28.3 preload ...
    > preloaded-images-k8s-v18-v1...:  341.16 MiB / 341.16 MiB  100.00% 61.07 M
🔥  Creating docker container (CPUs=2, Memory=4000MB) ...
🐳  Preparing Kubernetes v1.28.3 on Docker 24.0.7 ...
    ▪ Generating certificates and keys ...
    ▪ Booting up control plane ...
    ▪ Configuring RBAC rules ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
💡  kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
amitavaghosh@Amitavas-MBP addressbook-demo % alias kubectl="minikube kubectl --"
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get services
    > kubectl.sha256:  64 B / 64 B [-------------------------] 100.00% ? p/s 0s
    > kubectl:  52.80 MiB / 52.80 MiB [--------------] 100.00% 5.01 MiB p/s 11s
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   57s
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods -A
NAMESPACE     NAME                               READY   STATUS    RESTARTS      AGE
kube-system   coredns-5dd5756b68-c4pgq           1/1     Running   0             71s
kube-system   etcd-minikube                      1/1     Running   0             85s
kube-system   kube-apiserver-minikube            1/1     Running   0             85s
kube-system   kube-controller-manager-minikube   1/1     Running   0             85s
kube-system   kube-proxy-rzk22                   1/1     Running   0             71s
kube-system   kube-scheduler-minikube            1/1     Running   0             85s
kube-system   storage-provisioner                1/1     Running   1 (67s ago)   84s
amitavaghosh@Amitavas-MBP addressbook-demo %
```

3. Write a deployment.yml file to deploy addressbook application.

deployment.yaml
===============

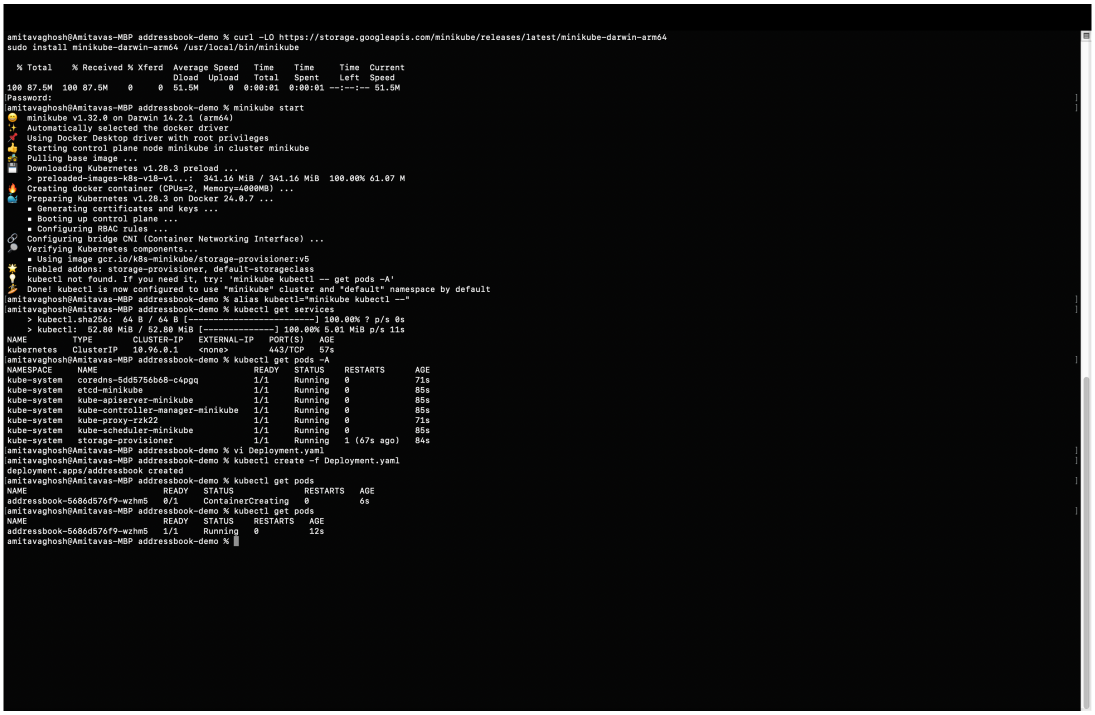
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: addressbook
  labels:
    app: addressbook
spec:
  replicas: 1
  selector:
    matchLabels:
      app: addressbook
  template:
    metadata:
      labels:
        app: addressbook
```

```
    spec:
      containers:
      - name: addressbook
        image: amitspring/addressbook-demo:3.0
        ports:
        - containerPort: 80
```



4. Create the deployment with deployment.yml.

kubectl create -f Deployment.yaml



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    ▪ Configuring RBAC rules ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
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    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
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NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
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amitavaghosh@Amitavas-MBP addressbook-demo % vi Deployment.yaml
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deployment.apps/addressbook created
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods
NAME                           READY   STATUS              RESTARTS   AGE
addressbook-5686d576f9-wzhm5   0/1     ContainerCreating   0          6s
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods
NAME                           READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-wzhm5   1/1     Running   0          12s
amitavaghosh@Amitavas-MBP addressbook-demo %
```

5. Scale up the addressbook application up to 3 instances.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: addressbook
  labels:
    app: addressbook
spec:
  replicas: 3
  selector:
    matchLabels:
      app: addressbook
  template:
    metadata:
      labels:
        app: addressbook
    spec:
      containers:
      - name: addressbook
        image: amitspring/addressbook:3.0
        ports:
        - containerPort: 80
```

```
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sudo install minikube-darwin-arm64 /usr/local/bin/minikube

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    ▪ Configuring RBAC rules ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
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amitavaghosh@Amitavas-MBP addressbook-demo % alias kubectl="minikube kubectl --"
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NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   57s
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NAMESPACE     NAME                               READY   STATUS    RESTARTS      AGE
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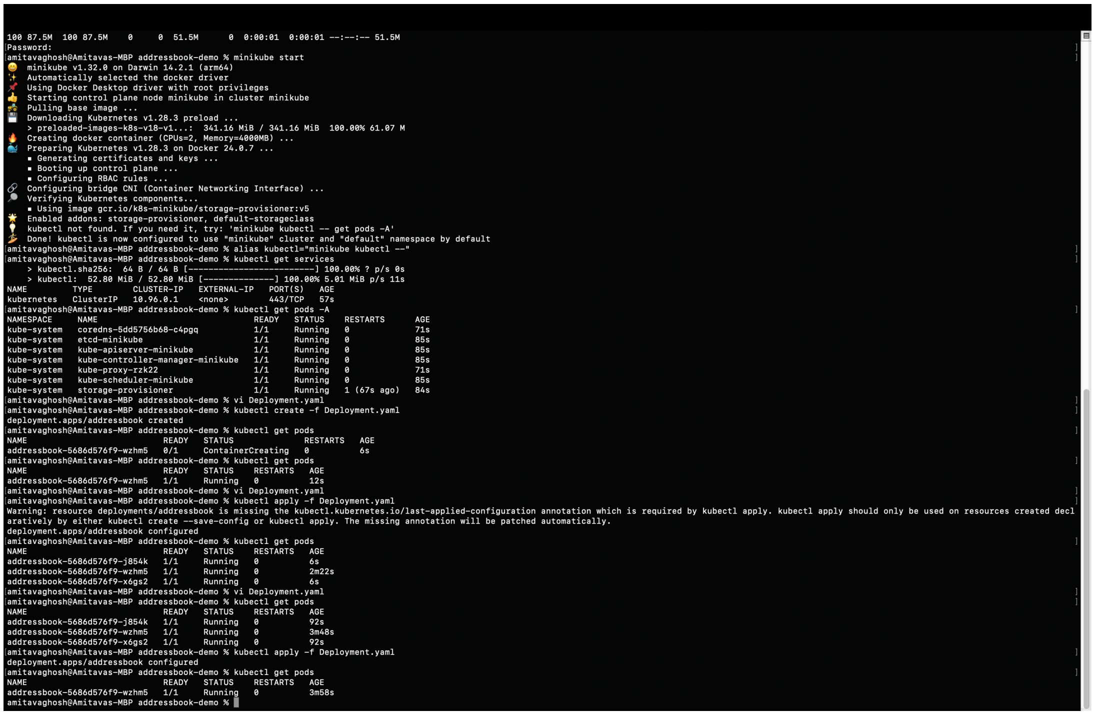
kube-system   kube-apiserver-minikube            1/1     Running   0             85s
kube-system   kube-controller-manager-minikube   1/1     Running   0             85s
kube-system   kube-proxy-rzk22                   1/1     Running   0             71s
kube-system   kube-scheduler-minikube            1/1     Running   0             85s
kube-system   storage-provisioner                1/1     Running   1 (67s ago)   84s
amitavaghosh@Amitavas-MBP addressbook-demo % vi Deployment.yaml
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl create -f Deployment.yaml
deployment.apps/addressbook created
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods
NAME                           READY   STATUS              RESTARTS   AGE
addressbook-5686d576f9-wzhm5   0/1     ContainerCreating   0          6s
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods
NAME                           READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-wzhm5   1/1     Running   0          12s
amitavaghosh@Amitavas-MBP addressbook-demo % vi Deployment.yaml
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl apply -f Deployment.yaml
Warning: resource deployments/addressbook is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created decl
aratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
deployment.apps/addressbook configured
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods
NAME                           READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k   1/1     Running   0          6s
addressbook-5686d576f9-wzhm5   1/1     Running   0          2m22s
addressbook-5686d576f9-x6gs2   1/1     Running   0          6s
amitavaghosh@Amitavas-MBP addressbook-demo %
```

6. Scale down the addressbook application to 1 instance.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: addressbook
  labels:
    app: addressbook
spec:
  replicas: 1
  selector:
    matchLabels:
      app: addressbook
  template:
    metadata:
      labels:
        app: addressbook
    spec:
      containers:
      - name: addressbook
        image: amitspring/addressbook:3.0
        ports:
        - containerPort: 80
```



7. expose your addressbook application deployment using a Kubernetes service.

kubectl expose deployment addressbook --type=NodePort --port=8080

```
Starting control plane node minikube in cluster minikube
Pulling base image ...
Downloading Kubernetes v1.28.3 preload ...
  > preloaded-images-k8s-v18-v1...:  341.16 MiB / 341.16 MiB  100.00% 61.07 M
Creating docker container (CPUs=2, Memory=4000MB) ...
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  ▪ Generating certificates and keys ...
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  ▪ Configuring RBAC rules ...
Configuring bridge CNI (Container Networking Interface) ...
Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: storage-provisioner, default-storageclass
kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
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amitavaghosh@Amitavas-MBP addressbook-demo % alias kubectl="minikube kubectl --"
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get services
    > kubectl.sha256:  64 B / 64 B [-------------------------] 100.00% ? p/s 0s
    > kubectl:  52.80 MiB / 52.80 MiB [--------------] 100.00% 5.01 MiB p/s 11s
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   57s
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kube-system   kube-controller-manager-minikube   1/1     Running   0             85s
kube-system   kube-proxy-rzk22                   1/1     Running   0             71s
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amitavaghosh@Amitavas-MBP addressbook-demo % vi Deployment.yaml
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl create -f Deployment.yaml
deployment.apps/addressbook created
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods
NAME                           READY   STATUS              RESTARTS   AGE
addressbook-5686d576f9-wzhm5   0/1     ContainerCreating   0          6s
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get pods
NAME                           READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-wzhm5   1/1     Running   0          12s
amitavaghosh@Amitavas-MBP addressbook-demo % vi Deployment.yaml
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl apply -f Deployment.yaml
Warning: resource deployments/addressbook is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created decl
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deployment.apps/addressbook configured
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NAME                           READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k   1/1     Running   0          6s
addressbook-5686d576f9-wzhm5   1/1     Running   0          2m22s
addressbook-5686d576f9-x6gs2   1/1     Running   0          6s
amitavaghosh@Amitavas-MBP addressbook-demo % vi Deployment.yaml
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NAME                           READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-j854k   1/1     Running   0          92s
addressbook-5686d576f9-wzhm5   1/1     Running   0          3m48s
addressbook-5686d576f9-x6gs2   1/1     Running   0          92s
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deployment.apps/addressbook configured
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NAME                           READY   STATUS    RESTARTS   AGE
addressbook-5686d576f9-wzhm5   1/1     Running   0          3m58s
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl expose deployment addressbook --type=NodePort --port=8080
service/addressbook exposed
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl get services
NAME          TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
addressbook   NodePort    10.109.209.9   <none>        8080:32285/TCP   11s
kubernetes    ClusterIP   10.96.0.1      <none>        443/TCP          9m54s
amitavaghosh@Amitavas-MBP addressbook-demo %
```

kubectl port-forward services/addressbook 7080:8080

```
  > preloaded-images-k8s-v18-v1...:  341.16 MiB / 341.16 MiB  100.00% 61.07 M
Creating docker container (CPUs=2, Memory=4000MB) ...
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service/addressbook exposed
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NAME          TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
addressbook   NodePort    10.109.209.9   <none>        8080:32285/TCP   11s
kubernetes    ClusterIP   10.96.0.1      <none>        443/TCP          9m54s
amitavaghosh@Amitavas-MBP addressbook-demo % kubectl port-forward services/addressbook 7080:8080
Forwarding from 127.0.0.1:7080 -> 8080
Forwarding from [::1]:7080 -> 8080
```

8. Verify your application.

Run the application on localhost with the port being forwarded I.e. 7080 in this case.